

Consumer-Relative Quantum Distinguishability on Causal Boundaries

Andrew H. Bond*

San José State University, San José, California, USA

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Distinguishability between quantum states is usually treated as a property of the states alone. We treat it instead as a property of the states together with a declared consumer, meaning the agent whose access channel, task, and budget determine which distinctions can be claimed. We define the consumer channel, the consumer-relative divergence, consumer equivalence, and task-relative distortion, verify that the resulting instruments reproduce exact classical anchors from an earlier projection-entropy campaign, and measure their behavior in a small spin-chain model where every quantity is computable. Three findings result. First, a nested hierarchy of consumers divides one fixed distinguishability total in a way that moves with the excitation, so a certificate built from one observable class can be exactly vacuous for an excitation in another class. Second, at scarce storage budgets an encoder optimized for a declared energy functional preserves that functional exactly while losing badly on global state fidelity, the encoder optimized for fidelity does the opposite, and the trade disappears once the budget is large enough to store everything the task needs. Third, a consistency requirement imposed across a family of consumers does no work when the consumers are probed disjointly and does real work, selecting the weak-coupling regime of the model family, exactly when all consumers witness one shared event. The second and third findings were subsequently confirmed under sealed preregistrations executed on scenario and coupling grids fully disjoint from the exploratory runs, with margins and thresholds fixed in advance. We also report three instrument defects found during construction and converted into standing controls, because each is a silent failure mode of restricted-divergence numerics. Everything numerical in this paper is demonstrated in finite-dimensional models. Nothing in it is a claim about physical gravity, real horizons, or quantum gravity, and the consumer does not create anything physical by choosing what to observe.

I. INTRODUCTION

Several modern derivations of gravitational field equations from information pass through a single kind of object, the distinguishability of quantum states as seen from a restricted vantage point. Jacobson derived the Einstein equation as an equation of state by demanding that a thermodynamic identity hold for every local causal horizon, which is a universal quantifier over restricted observers [7]. Bianconi postulated an action that is itself a quantum relative entropy between geometric and matter degrees of freedom [8]. Dorau and Much proved, for a special class of states on a special geometry, that the relative entropy accessible to an observer confined outside a horizon equals an energy flux, and recovered semiclassical gravity from that equality together with an assumed entropy-area formula [9]. A related construction runs the thermodynamic argument on stretched light cones [10]. In each case the restriction to an observer's accessible algebra is the informational move, and in each case the observer appears only as a restriction. None of these constructions contains a consumer in the operational sense, an agent with a declared task whose success is scored on that task rather than on full state reconstruction.

This paper supplies and tests that missing layer in the only setting where every quantity can be computed and audited, namely finite-dimensional models. The program it extends, called Observation Theory in our earlier classical work, holds that distinguishability is not one thing. It

is relative to a consumer, and the distinctions a consumer needs for its task can be preserved while global state fidelity is sacrificed, and conversely. The classical version of that statement was established with projection and entropy instruments in prior work of this campaign. Here the classical read operator is replaced by a quantum channel, the most general physically allowed map on quantum states, and the same questions are asked again.

Three claim types are kept separate throughout. Instrument claims state that the implemented objects compute what the definitions name, and they are verified against closed forms. In-model structural claims state measured facts about one declared family of finite-dimensional models. Physical claims about gravity, real horizons, or quantum gravity are absent, and no sentence of this paper should be read as making one. The causal direction in every construction runs from the physical state and the model couplings, through the accessible channel, to the consumer-relative divergence. No choice of channel, task, or budget feeds back into the state or the couplings. The consumer does not create anything physical by choosing what to observe. Observation relativity, as used here, concerns evidential access, not reality.

One further discipline governs the whole paper. Renaming known quantities adds nothing. Calling relative entropy distinguishability, calling the data-processing inequality an access law, and calling a partial trace a consumer channel are relabelings of standard mathematics. The contribution, if any, is confined to the task layer defined below and to the measured results. Where a result turns out to be a relabeling, the paper says so.

The paper proceeds in six steps. It replaces the clas-

* andrew.bond@sjsu.edu; ORCID 0009-0003-2599-6158

sical read operator with a quantum channel and states the access law. It defines consumer equivalence and task-relative distortion. It verifies exact classical reduction against anchors inherited from the earlier campaign. It analyzes a small transverse-field Ising model with a declared inside and outside, reporting instrument results and three defects converted to standing controls. It measures a consumer hierarchy, a budgeted encoding trade we call the flip, and a consumer-family consistency requirement with a two-sided outcome. It closes by stating precisely what would be required to connect any of this to the gravitational results cited above, and why the present constructions do not.

II. DEFINITIONS

A. Consumer channel and consumer-relative divergence

A quantum channel is a completely positive trace-preserving linear map $\mathcal{N}$ on density matrices [6]. A consumer C receives states only through its declared channel $\mathcal{N}_C$. Canonical instances used in this paper are the partial trace over an inaccessible tensor factor, local dephasing at declared strength, and pinching, meaning full dephasing in the computational basis, which restricts the consumer to the diagonal algebra.

The default divergence is the Umegaki quantum relative entropy [1],

$$D(\rho \parallel \sigma) = \text{Tr}[\rho (\ln \rho - \ln \sigma)], \quad (1)$$

in nats, the natural-logarithm unit of information, defined as $+\infty$ when the support of ρ is not contained in the support of σ . The consumer-relative divergence is

$$D_C(\rho \parallel \sigma) = D(\mathcal{N}_C(\rho) \parallel \mathcal{N}_C(\sigma)). \quad (2)$$

B. The access law

For every channel $\mathcal{N}$ and all states ρ, σ ,

$$D(\mathcal{N}(\rho) \parallel \mathcal{N}(\sigma)) \leq D(\rho \parallel \sigma), \quad (3)$$

the data-processing inequality [2]. Equality characterizes sufficiency of the channel for the pair [5]. In consumer language Eq. (3) says that claimable distinctions cannot exceed underlying distinctions, and that restriction, coarse-graining, and inaccessibility only erase distinguishability, never create it. This reading is a relabeling of a known theorem. The instruments of Sec. IV do not test whether Eq. (3) is true. They verify that the implementation computes the objects the theorem is about, which Sec. IV shows is a real failure surface.

C. Consumer equivalence

Let $\mathcal{A}_C$ be the algebra of observables the consumer can evaluate on the output of its channel. Two states are consumer equivalent,

$$\rho \sim_C \sigma \iff \text{Tr}(\rho O) = \text{Tr}(\sigma O) \text{ for all } O \in \mathcal{A}_C, \quad (4)$$

when no observable available to the consumer separates them. The quotient of state space by $\sim_C$ is the world the consumer can act on, with unreadable distinctions collapsed. In this paper the channel is the primitive object and $\mathcal{A}_C$ is derived from it, because two consumers with the same algebra can differ in noise, and a conditional expectation onto a subalgebra exists only for special algebra-state pairs.

D. Task-relative distortion

A task is a declared functional f on states, fixed before any sweep. For an encoding map $\mathcal{E}$ at a declared budget, with a declared decoder $\mathcal{D}$, the task distortion is the error in the functional,

$$d_f(\rho, \mathcal{E}) = |f(\rho) - f(\mathcal{D}(\mathcal{E}(\rho)))|, \quad (5)$$

scored against the global reconstruction error

$$1 - F(\rho, \mathcal{D}(\mathcal{E}(\rho))), \quad F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2, \quad (6)$$

where F is the Uhlmann fidelity [3, 4]. The task layer is where the paper's contribution lives, if anywhere. The divergence layer of Eqs. (1) through (4) answers how distinguishable two states are through a channel. The task layer asks which distinction a particular consumer needs, and the experiments below measure whether that question has content beyond the divergence layer.

E. Boundary language

Throughout, a boundary is a statement about which tensor factor of a finite-dimensional Hilbert space a consumer can touch, with the consumer channel a partial trace over the rest. A bipartite lattice cut is already a boundary in this sense. The causal boundaries of the gravitational literature are the special case in which the split is causal and the restricted state is thermal with respect to a modular flow, the intrinsic evolution that the state itself induces on the accessible algebra. The models of this paper use the general case and claim nothing about the special one. Section VIII lists what the special case would additionally require.

III. CLASSICAL REDUCTION AND ANCHORS

When all states and channels are simultaneously diagonalizable, the quantum objects must collapse to the

classical ones. Equation (1) becomes the Kullback-Leibler divergence, the channel becomes a stochastic map, and consumer equivalence becomes the classical quotient. The earlier campaign supplies two exact anchors, the branch-weight distributions of two fold projections whose entropies are known in closed form. A two-branch fold carries branch weights $(\frac{1}{2}, \frac{1}{2})$ and exactly one bit. A double fold carries branch weights $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ at its symmetric slice and exactly 1.5 bits.

The declared embedding, fixed before any run, maps fold branches in increasing order of the evolution parameter to computational basis states, coarea weights to eigenvalues of a diagonal density matrix, the uniform distribution to the reference state I/d , and the branch-label algebra to the diagonal algebra. Entropy is then recovered through the divergence pathway,

$$S(\rho) = \log_2 d - \frac{D(\rho \| I/d)}{\ln 2}. \quad (7)$$

The instrument reproduces 1.000000000000 bits for the two-branch fold and 1.500000000000 bits for the double fold through Eq. (7), at residuals below 2×10^{-15} .

The anchors also pin the equality case of the access law. Merging the two orientation branches of the double fold is a classical channel. Against the uniform reference the merge loses exactly nothing, because the conditional distributions inside the merged group coincide, so the data-processing gap is an exact sufficiency equality, measured at 1.1×10^{-16} . Against the asymmetric reference $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ the same merge loses a nonzero amount with the closed form

$$\Delta = \frac{\ln 2}{4} - \frac{1}{2} \ln \frac{4}{3} \approx 0.029446 \text{ nats}, \quad (8)$$

and the measured gap matches Eq. (8) to twelve decimals. A quantum instrument that failed these classical limits would be invalid regardless of anything else it computed. The embedding is declared rather than unique, and the reduction check is exactly as strong as that declaration.

IV. MODEL AND INSTRUMENT

A. Declared model

The model for all quantum runs is an open chain of 8 qubits with the transverse-field Ising Hamiltonian

$$H = -J \sum_{k=0}^6 Z_k Z_{k+1} - h \sum_{k=0}^7 X_k, \quad (9)$$

in the gapped paramagnetic regime. The reference state is thermal, $\sigma_\beta = e^{-\beta H} / \text{Tr } e^{-\beta H}$, at $\beta = 0.5$ with $J = 1$ and $h = 2$ for the instrument, hierarchy, and flip studies, and at $\beta = 0.4$ over the coupling grid $J \in \{0.6, 0.8, 1.0, 1.2\}$, $h \in \{1.6, 1.8, 2.0, 2.2\}$ for the family study of Sec. VII. Excitations are local unitaries $U_\theta = \exp(i\theta P_s)$ for a single-site Pauli operator P_s , applied to the thermal reference.

The declared cut places sites 0 through 3 inside and sites 4 through 7 outside, and the exterior consumer's channel is the partial trace onto the outside. All computations are exact diagonalization in double precision.

B. Instrument results

The instrument layer verifies that the implementation computes Eqs. (1) through (2) correctly. Closed-form checks cover the commuting Kullback-Leibler case, a pure state against the maximally mixed state, self-divergence, unitary invariance, and the support-violation convention, with a maximum residual below 2×10^{-15} . Across the excitation and channel sweep the implementation performed 32 data-processing checks of Eq. (3), every margin strictly positive and finite, minimum margin 9.6×10^{-8} . Composition monotonicity, meaning that composing two channels never increases the divergence, held at every point. Sweeping the numerical support floor across 10^{-14} through 10^{-10} changed no reported divergence, with a measured spread of zero.

C. Three defects converted to standing controls

Three instrument defects were discovered by failed first designs. Each is reported here as a methods result, because each is a silent failure mode of restricted-divergence numerics that produces passing-looking output, and each is now a standing control asserted on every run.

Silent infinite divergence at the support floor. At $\beta = 1$ the smallest thermal eigenvalue, near 2×10^{-12} , sat at the 10^{-12} support floor of the divergence routine. A full-rank reference was thereby misread as rank deficient, the full-state divergence went silently infinite, and every global-versus-restricted margin became infinity minus a finite number. All margins were vacuously positive, and the defect was invisible in the minimum margin. The production model must now clear the support floor by three decades, which is asserted, and every global divergence carries a finiteness assertion. The first evidence record contained this flaw and was regenerated. The corrected record supersedes it.

No-signalling null. A unitary, or any channel, acting only on the traced-out interior cannot change the exterior reduced state at all. The first sweep design placed the excitation inside, so every consumer divergence was exactly zero and the data-processing checks passed trivially on a degenerate sweep. Interior excitation is now an asserted null control, with the measured exterior divergence below 7×10^{-16} for excitations both deep inside and at the interior side of the cut.

Symmetry null. By the global spin-flip symmetry of Eq. (9), the single-site reduced thermal state is exactly $(I + mX)/2$ for a magnetization parameter m . An X -axis rotation commutes with it and is invisible to the site-only consumer even though it acts on that consumer's

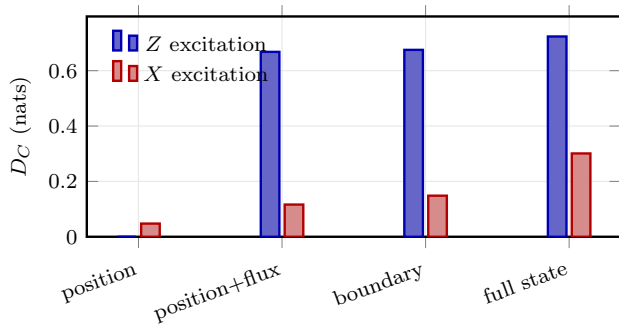


FIG. 1. Consumer hierarchy at excitation strength 0.8. For the Z excitation the position consumer sees exactly nothing and almost the whole distinguishability arrives at the flux rung. For the X excitation the position consumer already sees the event through correlations and the largest gap sits at the interior rung. The same four consumers divide the two distinguishability totals entirely differently.

own qubit, with the measured on-site divergence below 7×10^{-16} . Whether a consumer can see an excitation depends on the interplay of excitation direction and state symmetry, not only on locality. This null is also the first concrete instance in this paper of certificate vacuity, developed quantitatively in Sec. V.

V. CONSUMER HIERARCHY

A. Design

Four consumers are arranged as a degradation chain, each weaker consumer a channel composed onto the stronger one's output, so the ordering of their divergences is a theorem and any measured inversion would be an implementation bug. From weakest to strongest the levels are position only, realized as a Z -basis pinch on every outside site, position plus flux, which keeps the full single-site algebra of the excited site and pinches the rest, the full boundary, which is the partial trace onto the outside region, and the full state. The declared flux stand-in is the transverse-field energy density on the excited site,

$$f(\rho) = \text{Tr}[\rho(-h X_s)], \quad (10)$$

declared rather than derived, a choice whose risks are discussed with the results. Two excitation axes are measured, a Z -axis rotation and an X -axis rotation on the same outside site, both at $\theta \in \{0.4, 0.8, 1.2\}$.

B. Results

Table I shows the measured chains. For the Z -axis excitation the position-only consumer sees exactly nothing. This is a commutation theorem, asserted in the instrument, because the excitation commutes with every

TABLE I. Consumer-hierarchy divergence chains at excitation strength $\theta = 0.8$, in nats. Each level is a channel degradation of the one below it, so each column is nondecreasing downward by Eq. (3). The orderings and gap locations are stable across $\theta = 0.4$ and $\theta = 1.2$.

Consumer level	Z excitation	X excitation
position only	0.000000	0.047621
position plus flux	0.667985	0.116320
full boundary	0.675183	0.148405
full state	0.723532	0.300990

Z -string the pinched consumer can read. The largest gap, 0.667985 nats, sits at the flux rung. For the X -axis excitation the position-only consumer does see the event, at 0.047621 nats, because the excitation disturbs position correlations even though the on-site marginal is unchanged by the symmetry null of Sec. IV C. The largest gap for this excitation, 0.152585 nats, sits at the interior rung between the full boundary and the full state.

The structural statement is that the same four consumers, ordered the same way by Eq. (3), divide the same kind of distinguishability total entirely differently depending on the excitation sector. Where the distinguishing information lives moves with the excitation. A certificate of the form “the position consumer sees nothing, therefore nothing happened” is exactly vacuous for a flux-sector excitation, since the position consumer's divergence is identically zero there while the flux consumer's divergence is large. This is the task layer measured rather than relabeled. The divergence layer alone fixes only the ordering. The gaps, and where they sit, are model facts that no renaming produces. The caveat is adjacent. The flux functional Eq. (10) is a declared stand-in on a lattice with no null direction, and every statement above is a statement about this model class at these parameters.

VI. THE FLIP

A. Design

The flip hypothesis, transplanted from the classical program, is that at a fixed budget an encoder optimized to preserve a declared task functional beats a fidelity-optimized encoder on the task while losing on global fidelity. The declared design is as follows. The model is the outside region of Sec. IV, four qubits with the thermal marginal as vacuum. Scenario states apply a strong Z rotation on a fidelity-dominant site and a weak Z rotation on the flux site, so the fidelity signal and the task signal live on different qubits. The task is the flux functional Eq. (10) on the flux site. The encoder family, identical for every arm, keeps a subset of k of the 4 qubits and decodes by reattaching the vacuum marginals of the discarded qubits. The budget is the number of stored qubit slots.

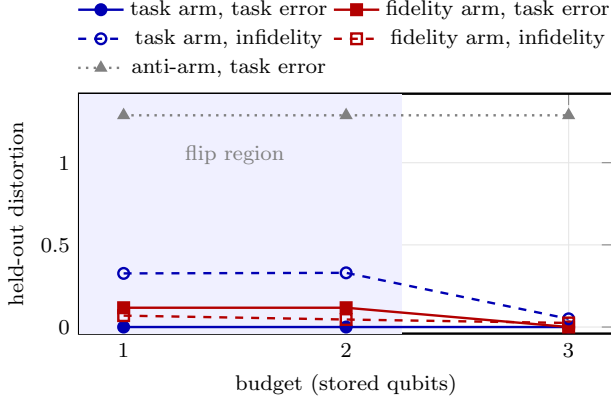


FIG. 2. The flip on held-out states. At budgets one and two the task arm preserves the flux functional exactly while paying a large infidelity price, and the fidelity arm does the opposite. At budget three the fidelity-optimal subset happens to contain the flux qubit and the trade disappears, so the flip is a scarcity phenomenon. The anti-arm is worst on the task at every budget, which certifies that the declared functional governs the task.

TABLE II. Held-out flip results by budget k in stored qubit slots. Task distortion is Eq. (5) for the flux functional, infidelity is Eq. (6). Task values of 0.000 are exact zeros of the encoding in exact arithmetic, measured at 10^{-16} .

k	Arm	Keeps	Task distortion	Infidelity
1	F	strong site	0.118	0.069
1	T	flux site	0.000	0.326
1	anti	flux site, pinched	1.289	0.431
2	F	strong pair	0.118	0.046
2	T	flux pair	0.000	0.330
2	anti	flux pair, pinched	1.289	0.430
3	F	includes flux site	0.000	0.025
3	T	includes flux site	0.000	0.050
3	anti	flux site pinched	1.289	0.197

The family is discrete and exhaustively searched, so there is no optimizer whose artifacts could manufacture the result. Arms are selected on a training grid of scenario angles and judged only on a disjoint held-out grid, with declared win margins of 0.01 on both axes. The F-arm minimizes mean infidelity, the T-arm minimizes mean task distortion, and a mandatory anti-arm applies the T-arm's encoder with the flux qubit dephased before storage, spending the identical budget while destroying the flux observable. If the anti-arm were not clearly worst on the task, the declared functional would not be what governs the task and the run would be uninterpretable rather than positive. A two-qubit analytic flip control, which no search could fake, is part of the accompanying test suite.

B. Results

Table II reports the held-out outcomes. At budget $k = 1$ the T-arm keeps the flux site and achieves task distortion exactly zero, because a kept single-site marginal determines the single-site functional exactly, at the price of infidelity 0.326. The F-arm keeps the strong site and achieves infidelity 0.069 at task distortion 0.118. The task-optimized encoder wins the task outright and is worse on global fidelity by a factor of 4.7. This is the flip. At $k = 2$ the split persists, and the fidelity arm's choice is informative in its own right, since it prefers the correlated strong pair over adding the weakly excited flux qubit. Fidelity favors preserving correlation structure. At $k = 3$ the fidelity-optimal subset happens to contain the flux qubit, both arms reach task distortion zero, and the flip vanishes. The anti-arm is worst on the task at every budget, 1.289 against at most 0.118, so the declared functional does govern the task and the runs are interpretable.

The flip region is contiguous, budgets 1 and 2, and the flip dies at exactly the budget where storage suffices to keep everything task-relevant. The flip is a scarcity phenomenon. This is the budget-relativity structure the classical program predicts, here realized with quantum states, a quantum channel budget, and an energy functional. The result is not an estimation protocol. Classical shadow methods predict many observables of a state from few randomized measurements with guarantees that sharpen as the observable class narrows [11]. The flip instead measures a strict trade between one declared functional and global reconstruction at matched storage, and its content is the existence and location of that trade, not an estimator. Two caveats are adjacent. The scenario family was declared so that task signal and fidelity signal live on different qubits, which is the misalignment the hypothesis requires, so the result demonstrates existence in a declared model class and nothing stronger. The exploratory runs above were then subjected to a sealed preregistered replication, registration PREREG-QO2-001, whose scenario grids are fully disjoint from every angle used above and whose win margins of 0.01 on both axes were fixed before execution. The governed run confirms the flip at the scarce budget at the primary coupling point, with task distortions 0.0000 against 0.1076 and infidelities 0.3637 against 0.0678, and at both preregistered secondary coupling points, with the anti-arm worst on the task everywhere. Within this model class and encoder family the flip therefore carries the demonstrated-in-model label; any claim beyond this class would require new preregistration.

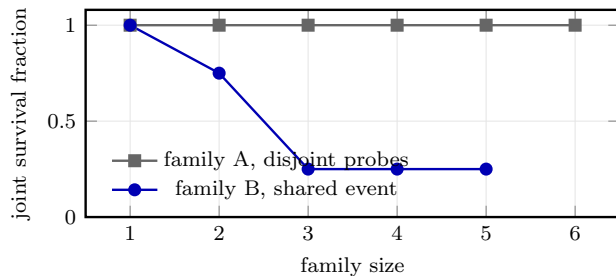


FIG. 3. Fraction of the coupling grid on which the family requirement holds, as consumers are added. For disjoint probes the joint requirement never removes a coupling point that a single consumer keeps, so the curve stays at one and the quantifier is inert. For nested windows witnessing one shared event the joint requirement tightens to one quarter of the grid, strictly below every single-consumer value, and the surviving quarter is exactly the weak-coupling row.

VII. CONSUMER-FAMILY CONSISTENCY

A. Design

Jacobson’s derivation runs on a universal quantifier, a demand imposed for every local horizon at once [7]. The model-internal question tested here is whether such a quantifier does any work, meaning whether a requirement imposed across a whole family of consumers constrains a model more than the same requirement imposed on each consumer alone. The declared requirement, per coupling point (J, h) of the model family at $\beta = 0.4$, is that each consumer’s divergence follow the declared flux weight,

$$D_c(\theta) = \lambda_c \theta^2 \quad \text{within relative residual } \epsilon, \quad (11)$$

with the constant λ_c universal across the family within spread δ , where the spread is the range of the constants divided by their mean. The flux weight θ^2 is declared from the excitation parameter alone, so nothing informational enters the declaration and the universality content is measured rather than inserted. Thresholds $\epsilon = \delta = 0.05$ were fixed with the design, and all verdicts are also reported at a loose value 0.15. Excitation strengths are $\theta \in \{0.05, 0.10, 0.15, 0.20\}$.

Two families are measured. Family A consists of six wedge consumers, the right exteriors of cuts 1 through 6, each probed by a Z rotation at its own boundary site, so the probes are disjoint. Family B consists of five nested windows, from the single shared site up to the full chain, all witnessing one shared excitation at one site, so every consumer sees the same event with a different extent of access.

B. Family A, disjoint probes

The quantifier is inert. The θ^2 law holds for every consumer at every coupling point, the constants λ_c agree

TABLE III. Family B universality spread of the constants λ across the five nested windows, by coupling point. The strict threshold is $\delta = 0.05$. The spread increases monotonically with the coupling ratio J/h , which sets the correlation strength.

	$h = 1.6$	$h = 1.8$	$h = 2.0$	$h = 2.2$
$J = 0.6$	0.0350	0.0345	0.0340	0.0335
$J = 0.8$	0.0612	0.0603	0.0594	0.0585
$J = 1.0$	0.0936	0.0922	0.0908	0.0894
$J = 1.2$	0.1313	0.1293	0.1273	0.1253

across the six consumers within a relative spread of at most 1.6×10^{-3} over the whole grid, more than an order of magnitude inside the declared δ , and the joint requirement never removes a coupling point that any single consumer keeps. Survival is 1.0 for every consumer and for every subfamily at both thresholds. The mechanism is visible in the construction. At short correlation length the effect of a boundary-local probe is confined near its site up to tails set by that length, every wedge containing its own probe captures it essentially completely, and disjoint probes tie nothing together. A family of observers probed disjointly cannot make an all-observers quantifier do work.

C. Family B, one shared event

The quantifier is active. Every window passes the θ^2 law singly at every coupling point, so no single consumer constrains the coupling grid at all. The joint universality requirement tightens strictly with family size, with surviving fractions of the sixteen-point coupling grid of 1.00, 0.75, and 0.25 at family sizes one, two, and three, and 0.25 thereafter. The joint requirement is strictly tighter than every single-consumer requirement. The surviving quarter is exactly the weak-coupling row $J = 0.6$. Table III shows why. The universality spread orders monotonically with correlation strength, from 0.0335 at the weakest-coupling corner to 0.1313 at the strongest-coupling corner, so the family requirement constrains couplings by bounding correlation assistance and selects the near-factorized regime. The constants also saturate once the window clears the correlation length. On the surviving row the three-site window and every larger window agree to four decimals, while at the strongest coupling that agreement is reached only by the two largest windows. All sufficiently large observers agree, and only observers smaller than the correlation length dissent.

D. Reading

Stated carefully and model-internally, an all-observers consistency requirement does real constraining work precisely when the observers overlap on shared events, and does none when they are probed disjointly. The overlap

structure is what the continuum family of local horizons in Jacobson’s argument possesses and what family A lacks. This is a statement about one finite model family and about the structure of the quantifier, not about gravity. Two caveats are adjacent. The verdict is threshold-relative, since at the loose threshold of 0.15 every coupling point survives every requirement, so a selection-effect claim requires a preregistered spread threshold. And in one dimension every cut has a boundary of a single site, so this experiment tests the universality content of the quantifier and says nothing about area scaling.

The preregistered study exists. Registration PREREG-QO3-001 fixed the spread threshold at 0.05 before execution and moved to a coupling grid and excitation grid fully disjoint from those above, stating activation and inertness as separately refutable claims. The governed run passes both. The joint shared-event requirement holds on one quarter of the fresh grid against single-window survivals of one, the disjoint-probe requirement adds exactly nothing beyond its tightest single consumer, and the surviving set is exactly the weak-coupling row of the fresh grid, reproducing the predicted secondary structure. Within this model family both claims therefore carry the demonstrated-in-model label.

VIII. WHAT A CONNECTION TO GRAVITY WOULD REQUIRE

This section states precisely what separates the constructions above from the gravitational results of Refs. [7, 9], so that no reader mistakes the finite models for more than they are.

Every algebra in this paper is a finite-dimensional type I factor. The observable algebra of a real horizon exterior is a type III von Neumann factor, which admits no density matrices, no partial trace, and no von Neumann entropy of the restricted state. Connecting the consumer layer to that setting would require, at minimum, four replacements. First, type III algebras in place of finite factors, with relative entropy defined through relative modular operators rather than Eq. (1). Second, a genuine causal structure in place of a lattice cut, since the cut used here has no light cones, no Killing flow, and no notion of a horizon generator. Third, the entropy-area input justified rather than declared, because in Refs. [7, 9] the proportionality of entropy to area is an assumption, and nothing in this paper produces or replaces it. Fourth, the special state class relaxed, since the theorem of Ref. [9] holds for coherent excitations on a bifurcate Killing horizon and the thermal lattice states used here have no counterpart standing.

The finite-dimensional statements divide accordingly.

The existence of reduced density matrices, of the pinching channel, and of Eq. (7) are artifacts of type I, and every number in Secs. IV through VII depends on them. The relative entropy itself and the data-processing inequality have exact type III counterparts, so the definitions of Sec. II survive the passage in principle. Whether the task layer survives it, meaning whether consumer equivalence for declared functionals and the flip can even be formulated without density matrices, is open, and this paper takes no position on it.

IX. CONCLUSION

This paper replaced the classical read operator of an earlier observation-theoretic program with a quantum channel, defined consumer-relative divergence, consumer equivalence, and task-relative distortion, and measured all of them in finite-dimensional models with exact classical anchors. The instrument layer reproduced closed forms at the 10^{-15} level, held every data-processing margin strictly positive, and yielded three transferable methods findings, the support-floor infinity, the no-signalling null, and the symmetry null, each a silent failure mode of restricted-divergence numerics now held as a standing control. The hierarchy result showed that where distinguishability lives moves with the excitation sector, making some certificates exactly vacuous. The flip result showed a strict task-versus-fidelity trade that exists at scarce budgets and dies when scarcity ends. The family result showed that an all-observers requirement constrains a model exactly when the observers share events, and is otherwise inert.

The relabeling clause applies to everything else. Whatever in this paper merely renames relative entropy, data processing, or the partial trace is not a contribution. The contribution is confined to the task layer and to the measured results, all of which are demonstrated in finite-dimensional models. The flip and the consumer-family results were confirmed under sealed preregistrations on grids disjoint from their exploratory runs and carry the demonstrated-in-model label for their declared model classes; the remaining results are exploratory. Nothing in the paper claims anything about physical gravity, real horizons, or quantum gravity. The consumer does not create anything physical by choosing what to observe.

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